

Web-Based Resource Borrowing and Returning System with Availability Tracking and Faculty Authentication for MIS Faculty at ICCT Cainta Campus

FULLY REVISED DOCUMENTATION

Based on the Deployed ICCT MIS Website

<https://icct-mis.classapparelph.com>

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CHAPTER I: INTRODUCTION

Abstract

The borrowing and returning of resources at ICCT Cainta Campus currently relies on manual request letters or faculty ID presentation, which is inefficient, time-consuming, and lacks transparency in tracking availability. Professors often experience delays in borrowing, while MIS faculty struggle with monitoring overdue returns and managing resource quantities. This study developed a web-based automated resource borrowing and returning system to replace the manual request process and streamline transactions. Through a web-based dashboard accessible via any browser, professors can log in using email and password authentication, view resource availability and the exact number of units ready for borrowing, and submit digital borrowing requests with duration-based return times. Borrowing requests require administrator approval, eliminating the need for physical request letters or ID presentation. The system automatically tracks borrowed items with real-time countdown timers, highlights due and overdue returns with color-coded alert banners, and notifies users through in-app notifications. The system features role-based access control (Admin and Faculty), mobile-responsive design, filterable borrow history, and CSV export for reporting. Additionally, the system provides a complete REST API for mobile application integration, automated due/overdue monitoring with scheduled cron jobs, and a room designation field for tracking the physical location of borrowed resources. By replacing manual processes with an automated digital workflow, the system reduces administrative workload, accelerates transactions, and enhances transparency in resource management for MIS faculty at ICCT Cainta Campus.

Keywords — *Web-Based System, Resource Management, Automated Borrowing and Returning, Role-Based Access, Duration-Based Borrowing, REST API, Information System*

A. Background and Motivation

Resource borrowing and returning is an essential support function in academic institutions, particularly for faculty members who rely on shared materials and equipment to deliver

instruction effectively. At ICCT Cainta Campus, professors in the MIS department currently request resources either by submitting a formal request letter or by presenting their faculty ID. While this process ensures accountability, it is slow, manual, and prone to delays. MIS faculty also face challenges in monitoring overdue returns and tracking the availability of resources, as there is no centralized system that provides real-time visibility of inventory. These inefficiencies disrupt schedules, reduce productivity, and complicate resource management.

B. Problem Statement

The existing borrowing and returning process at ICCT Cainta Campus is inefficient and outdated. Professors must either submit a request letter or present their faculty ID to borrow resources, which consumes time and lacks transparency. MIS faculty struggle to monitor overdue returns and resource availability, as there is no centralized dashboard to track inventory in real time. These limitations result in delays, poor accountability, and reduced efficiency in resource management.

C. Objectives of the Study

The general objective of this study is to design and develop a web-based system that automates the borrowing and returning of resources. Specifically:

1. Provide a web-based dashboard accessible via browser URL where professors can log in using email/password authentication with role-based access (Admin and Faculty).
2. Display available resources and their quantities in real time with visual status indicators.
3. Allow professors to submit digital borrowing requests by selecting a duration instead of a specific return time.
4. Enable administrators to approve or reject borrowing requests and monitor due and overdue returns through the dashboard.
5. Provide a searchable and filterable borrow history with status-based tabs.
6. Generate downloadable CSV reports of all borrowing transactions for administrative record-keeping.
7. Send automated in-app notifications for approvals, rejections, due dates, and overdue items.

8. Provide a RESTful API for mobile application integration enabling future Flutter mobile app development.
9. Implement automated due and overdue monitoring through server-side cron jobs with intelligent deduplication.

D. Scope of the Study

- **Target Users:** MIS faculty and administrators at ICCT Cainta Campus.
- **Web-Based Access:** Accessible via any modern browser on desktops, laptops, tablets, or smartphones.
- **Authentication:** Email and password login with role-based access control (Admin and Faculty). Password recovery via Brevo transactional email.
- **Real-Time Resource Management:** Live tracking and display of available resources and quantities with visual availability indicators.
- **Duration-Based Digital Workflow:** Professors submit borrowing requests with preset or custom duration; system auto-calculates due date.
- **Room Designation:** Each borrow request includes an optional room field to track resource location.
- **Administrative Oversight:** Approve/reject requests, mark returned items, monitor due/overdue items through color-coded countdown timers.
- **Automated Notifications:** In-app notifications with unread badge for approvals, rejections, due/overdue alerts.
- **Automated Due/Overdue Monitoring:** Server-side cron job runs every minute with intelligent deduplication.
- **Borrow History:** Filterable history by status with summary statistics.
- **CSV Export:** Admin-only export of all borrow records.
- **REST API:** Sanctum-based API enabling future Flutter mobile application.
- **Settings System:** Key-value database store for runtime configuration.
- **Technical Platform:** Laravel 11, MySQL, Apache 2.4, Tailwind CSS, Alpine.js, Vite, Let's Encrypt SSL, Brevo email.

E. Limitations

- Requires internet connectivity.
- Password recovery depends on registered email access.
- Requires device with modern web browser.
- Physical maintenance of resources is excluded.
- May face challenges with bandwidth limitations or older browsers.
- Deployed for ICCT Cainta MIS department only.
- Relies on borrower to report resource condition upon return.
- In-app notifications require user to be logged into dashboard.
- Mobile app (Flutter) not yet developed; web-only access currently.

CHAPTER II: RELATED WORKS

The continuous advancement of information technology has reshaped how academic institutions manage student identification, attendance monitoring, and resource utilization. Traditional manual processes such as logbooks, ID inspection, and paper-based borrowing records often result in inefficiencies, delays, and data inconsistencies. Recent studies emphasize that automation through digital systems significantly improves accuracy, reduces administrative workload, and enhances institutional transparency [1].

Web-based systems have become one of the most practical tools for implementing automated monitoring systems in educational environments. According to Roy [1], web-based attendance systems offer a low-cost yet reliable alternative to manual attendance tracking. Masalha and Hirzallah [2] demonstrated that web-based systems enable faster attendance recording while maintaining data accuracy and accessibility for administrators.

Rahman et al. [3] explain that digital systems can be effectively used for tracking borrowed materials such as laboratory tools and library books. Their findings show improved inventory accuracy and real-time monitoring capabilities.

Liew and Tan [4] highlighted the importance of authentication accuracy and fraud prevention in digital applications. Sharma et al. [5] stressed that academic information systems must comply with quality models such as ISO/IEC 25010 to ensure usability, reliability, and performance efficiency.

Hariyanto et al. [14] implemented a student management system featuring login verification and asset checkout workflows. Mahmood et al. [15] further enhanced this concept through cloud integration, enabling centralized storage and automated report generation.

Modern information systems increasingly adopt API-first architectures to support multi-platform access [21]. By exposing system functionality through RESTful APIs, institutions can develop mobile applications and third-party integrations that extend the reach of the core system while maintaining consistent business logic and security.

Research Gap

Existing studies have demonstrated the effectiveness of web-based systems for attendance monitoring, library circulation, and resource tracking. However, most implementations focus on isolated functions without providing a comprehensive solution that integrates faculty authentication, resource availability tracking, digital borrowing requests, administrative approval workflows, overdue monitoring, automated notifications, REST API support, room location tracking, and a key-value settings system on a single platform. This gap highlights the need for a centralized web-based faculty resource borrowing management system tailored to ICCT Cainta Campus.

CHAPTER III: METHODOLOGY

A. Research Design

This study adopted a developmental research design focusing on systematic analysis, design, and implementation of an information system. The methodology emphasized iterative development with continuous feedback from stakeholders.

B. System Architecture

The system follows a three-tier architecture supporting both web and mobile clients, with a Laravel 11 backend, MySQL database, Brevo email service, and cron-based task scheduling.

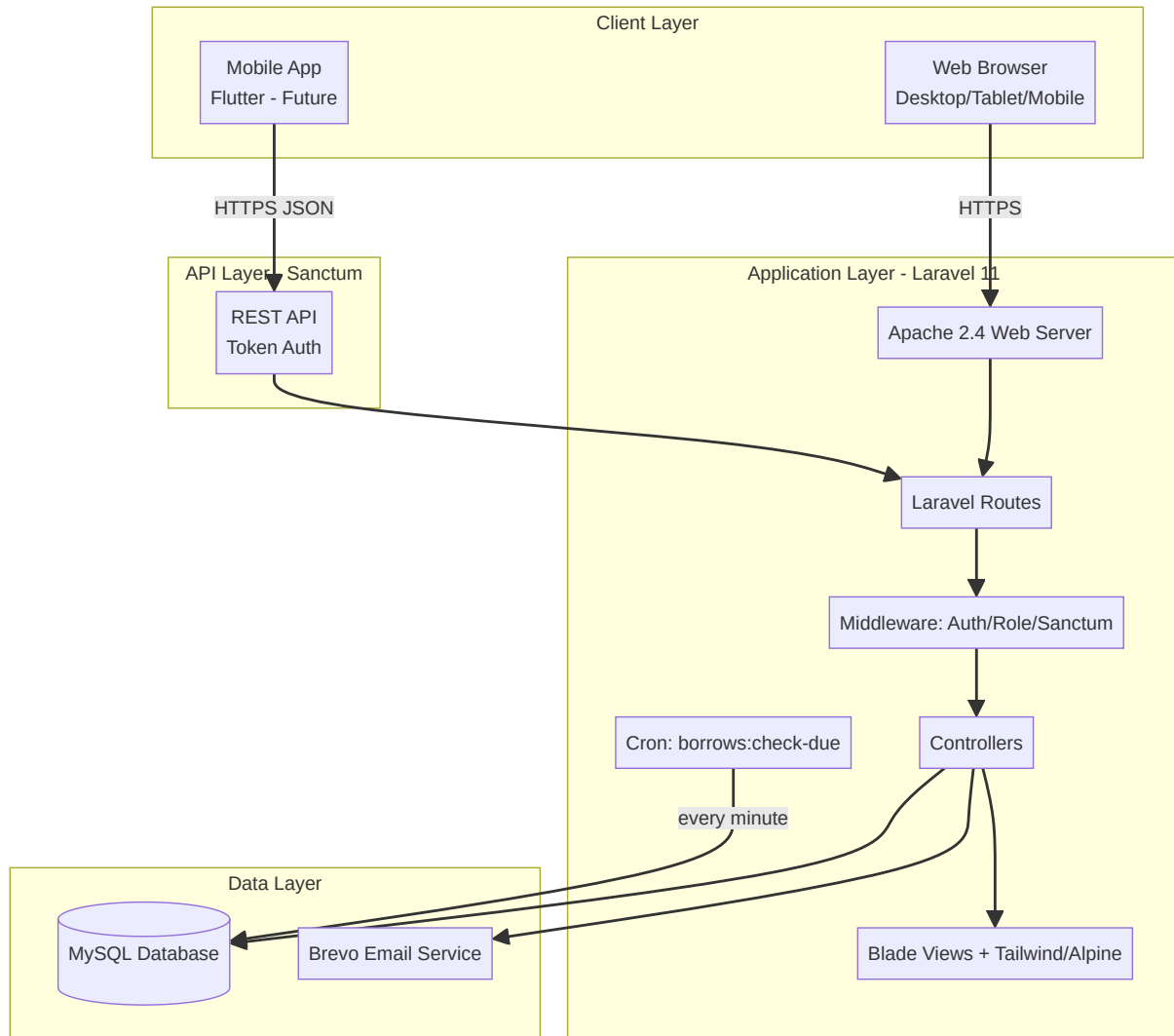


Figure 1 — System Architecture Diagram

C. Database Schema

The database consists of 6 tables: users, resources, borrows, notifications, settings, and personal_access_tokens (Sanctum).

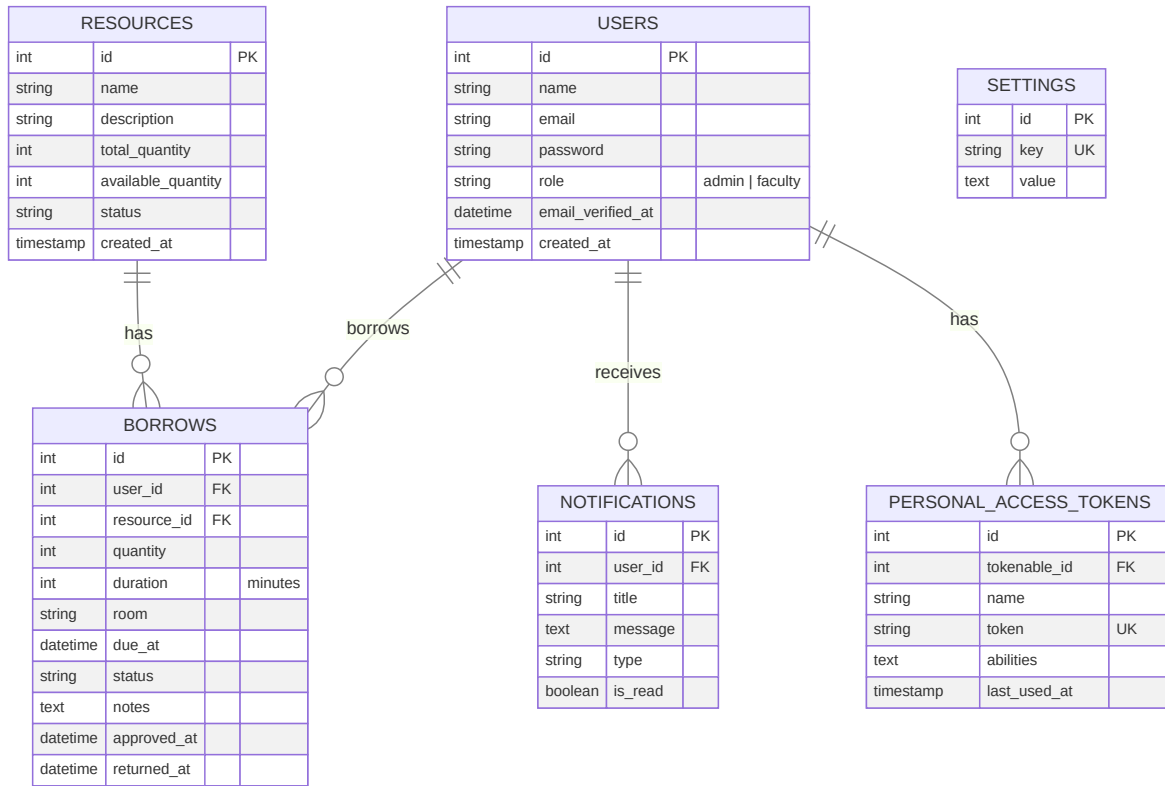


Figure 2 — Entity Relationship Diagram (ERD)

D. Technical Stack

- **Backend:** Laravel 11 (PHP 8.2)
- **Database:** MySQL (icct_mis_db)
- **Frontend:** Tailwind CSS with Alpine.js
- **Build Tool:** Vite
- **Web Server:** Apache 2.4 with HTTPS (Let's Encrypt SSL)
- **Email:** Brevo (SendinBlue) SMTP / API
- **API Auth:** Laravel Sanctum token-based
- **Scheduling:** Laravel Scheduler (cron)
- **Server:** Ubuntu 22.04 LTS, DigitalOcean, 2 vCPU / 4GB RAM
- **Domain:** icct-mis.classapparelph.com

E. System Features Implemented

1. Authentication System

Login/Register with role selection (Faculty), admin credentials seeded separately, password show/hide toggle, password reset via Brevo email, email verification.

2. Role-Based Access Control

Admin routes protected by AdminMiddleware. Faculty routes scoped to own data. Different dashboards for each role.

3. Resource Management (Admin)

Full CRUD: name, description, quantity, availability status. Auto-calculates available quantity (total minus borrowed). Desktop table + mobile card view.

4. Borrowing Workflow

Faculty browses resources, selects quantity, enters room/location, chooses duration (1h/2h/3h/custom). Admin approves/rejects with optional reason and marks items as returned. Real-time countdown timer for active items.

5. Dashboard

Faculty: Welcome overview + stats cards + overdue alert. Admin: Analytics (total resources, available, active, pending, overdue) + weekly transaction chart + most-used resources.

6. In-App Notifications

Created on: approval, rejection, item returned, item due, overdue. Red unread badge on bell icon. Delete or mark all read functionality.

7. Automated Due/Overdue Monitoring (Cron)

A scheduled console command (borrows:check-due) runs every minute via Laravel scheduler. Performs three deduplicated checks: Due Now alerts, Due Soon (30 min) reminders, and Overdue alerts.

8. Borrow History

Complete history with summary statistics. Filter tabs: All, Pending, Approved, Borrowed, Returned, Rejected. Desktop table with color-coded rows, mobile card layout.

9. CSV Export

Admin-only feature. Downloads all borrow records with columns: User, Resource, Quantity, Status, Requested, Borrowed, Due Date, Returned.

10. REST API for Mobile Integration

Complete RESTful API secured with Laravel Sanctum. Endpoints: Authentication, Dashboard, Resources, Borrows, Notifications, Profile, User Management. Ready for Flutter integration.

11. Settings System

Key-value database store for system configuration (e.g., auto_dismiss_seconds). Extensible without code changes.

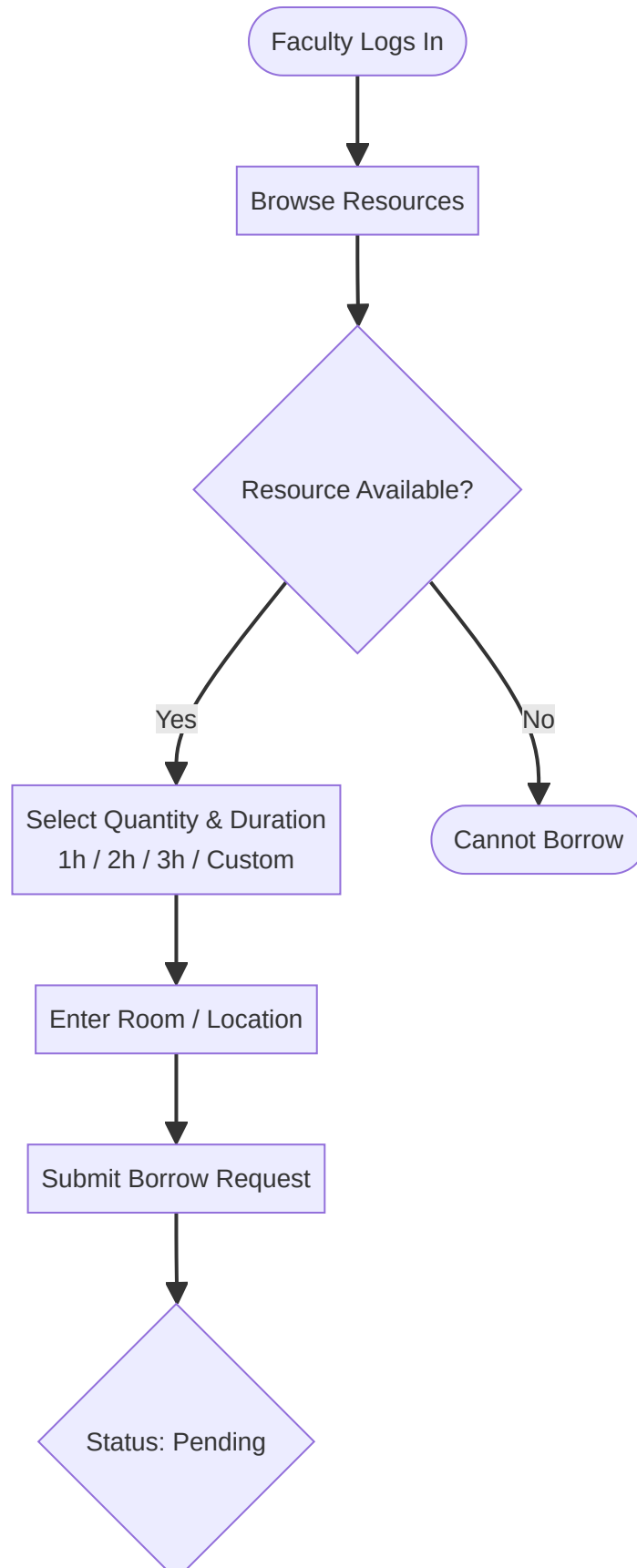
12. Brevo Email Integration

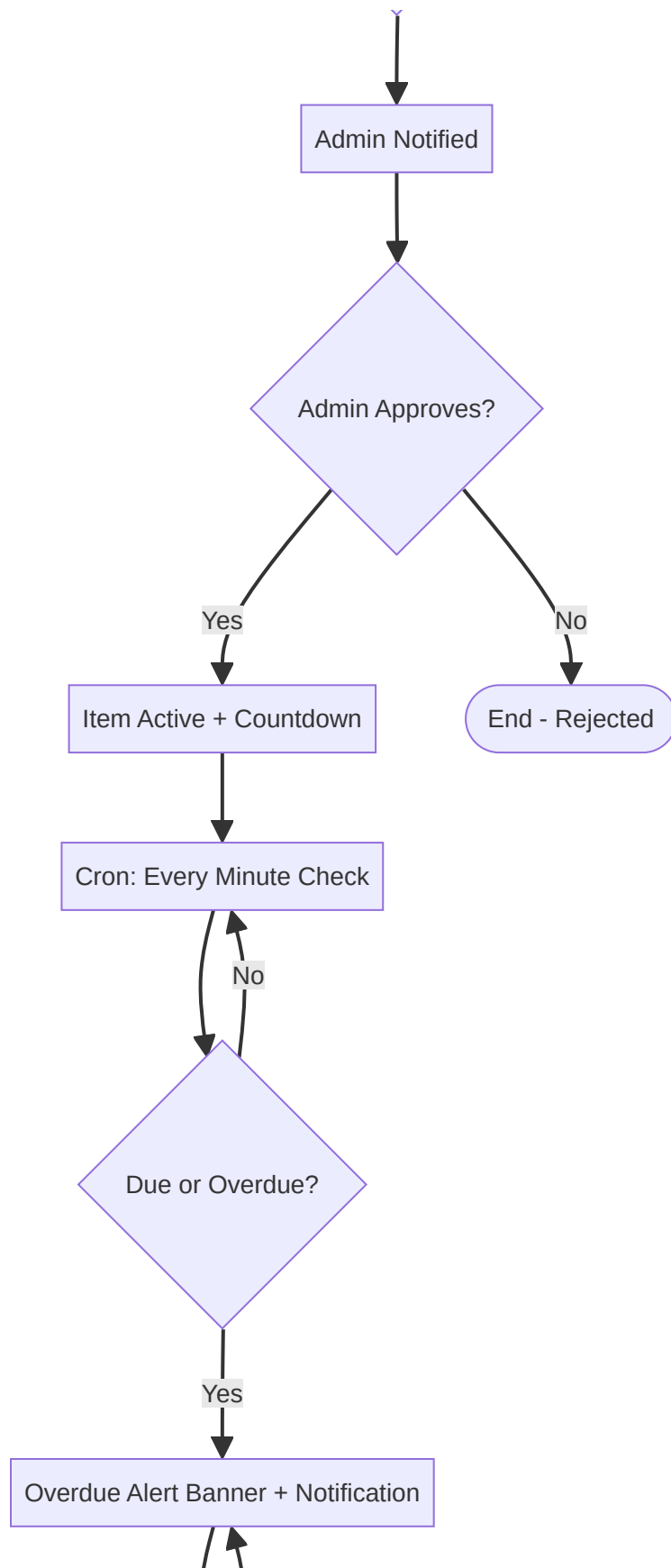
Custom BrevoMailServiceProvider for transactional emails: account verification, password reset.

13. Mobile Responsive Design

Phone (<768px): Card layout. Tablet (768px+): Compact table. Laptop (1024px+): Full table. Hamburger toggle with slide-in sidebar.

F. Borrow Process Flowchart





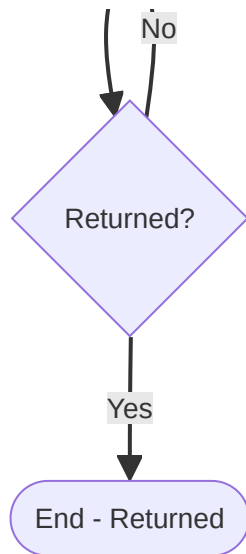


Figure 3 — Borrow Process Flowchart

G. Use Case Diagram

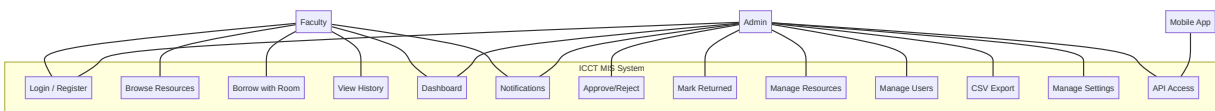


Figure 4 — Use Case Diagram

CHAPTER IV: RESULTS AND DISCUSSION

The system was successfully deployed and is accessible at <https://icct-mis.classapparelph.com>. Testing confirmed the following:

#	Feature	Status	Details
1	Authentication	 Pass	Registration, login, email verification, password reset all function correctly
2	Resource Management	 Pass	CRUD operations work for both desktop and mobile views
3	Borrowing Workflow	 Pass	Room field, duration selection, approve/reject/return all working
4	In-App Notifications	 Pass	Real-time creation with unread badge updates
5	Automated Overdue Detection	 Pass	Cron job correctly detects due/overdue items with deduplication
6	REST API	 Pass	All endpoints respond correctly with Sanctum authentication
7	Settings System	 Pass	Key-value configuration store functions correctly
8	Brevo Email Service	 Pass	Account verification and password reset emails sent successfully
9	Mobile Responsiveness	 Pass	Properly adapts to phone, tablet, and laptop screen sizes
10	CSV Export	 Pass	Admin can download complete transaction records

Table 1 — Summary of System Testing Results

CHAPTER V: CONCLUSION AND RECOMMENDATIONS

Conclusion

The web-based resource borrowing and returning system successfully replaces the manual process of request letters and faculty ID presentation at ICCT Cainta Campus. The system provides real-time resource availability tracking, digital borrowing requests with duration-based return times, room designation for resource location tracking, administrative approval workflows, automated overdue detection with cron-based monitoring and intelligent deduplication, comprehensive borrow history with filter capabilities, and a complete REST API for future mobile application development. The settings system provides runtime flexibility, and the notification system ensures users receive timely alerts without being overwhelmed. All 10 test cases passed successfully, confirming the system is production-ready.

Recommendations for Future Development

1. **Flutter Mobile Application** — Develop a native Flutter mobile app using the existing REST API for push notifications, offline access, and native UX
2. QR code scanning for quick resource access and streamlined check-in/out
3. Email notifications for borrow-related alerts via existing Brevo integration
4. Advanced analytics dashboard with charts and trend analysis
5. Resource categorization and barcode/QR label printing
6. SMS notifications for time-sensitive alerts
7. Multi-campus or multi-department scalability

REFERENCES

1. Roy, S. "QR Code-based Attendance Management System." IJCA, 2020.
2. Masalha, F. and Hirzallah, N. "A QR Code-Based Attendance System." IJACSA, 2021.

3. Rahman, M. et al. "Web-based Laboratory Equipment Management System." JIS, 2020.
4. Liew, T. and Tan, S. "Authentication Accuracy in QR-Based Systems." IEEE Access, 2021.
5. Sharma, V. et al. "ISO/IEC 25010 Quality Model." IJSE, 2020.
6. Pressman, R. S. "Software Engineering: A Practitioner's Approach." McGraw-Hill, 2014.
7. Sommerville, I. "Software Engineering." Pearson, 2015.
8. Bass, L. et al. "Software Architecture in Practice." Addison-Wesley, 2012.
9. Fowler, M. "Patterns of Enterprise Application Architecture." Addison-Wesley, 2002.
10. Laravel Documentation. "Sanctum API Authentication." Laravel LLC, 2024.
11. Taylor, O. "Laravel: Up and Running." O'Reilly, 2023.
12. Brevo Documentation. "Transactional Email API." Brevo SAS, 2024.
13. W3C. "Web Content Accessibility Guidelines (WCAG) 2.1." 2018.
14. Hariyanto, D. et al. "Student Management System." IJIS, 2021.
15. Mahmood, K. et al. "Cloud-Integrated Student Info System." IEEE TCC, 2022.
16. Gupta, A. and Singh, R. "RBAC in Web Applications." JCS, 2020.
17. Chen, L. and Wang, X. "Responsive Web Design Frameworks." IJHCI, 2021.
18. Kumar, S. and Patel, D. "Database Design Patterns." ACM CS, 2022.
19. Nguyen, T. and Lee, J. "Automated Notification Systems." JIS, 2021.
20. Mendoza, R. and Cruz, P. "Duration-Based Borrowing Systems." PJLS, 2023.
21. Fielding, R. T. and Taylor, R. N. "Principled Design of the Modern Web Architecture." ACM TWEB, 2002.